



# Software Development Process

# BIS605

# Software Development Technologies for Digital Business

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# References

- Murali Chemuturi, Requirements Engineering and Management for Software Development Projects, Springer, 2013
- Cecilia R. Aragon, Website Design, IEOR 170, UC Berkeley, Spring 2006
- <https://zurb.com/blog/response-times-the-3-important-limits>
- <https://medium.com/omarelgabrys-blog/software-engineering-process-activities-part-3-ca1ef6818fd6>
- <https://www.tutorialspoint.com/sdlc>
- [https://www.tutorialspoint.com/adaptive\\_software\\_development/sdlc\\_iterative\\_incremental\\_model.htm](https://www.tutorialspoint.com/adaptive_software_development/sdlc_iterative_incremental_model.htm)
- Resources from previous BIS605 Lectures

# **SOFTWARE DEVELOPMENT LIFECYCLE**

# Software Development Process

- Software development process commonly consists of the following activities:
  - Requirement gathering and analysis
  - Software design
  - Construction
  - Testing

# Requirement Gathering and Analysis

- Focuses on user requirements
  - To collect user requirement
  - To understand user requirement
  - To analyze user requirement
- Requirements can be classified based on three considerations
  - Functionality considerations
  - Product construction considerations
  - Source considerations

# Requirement Gathering

- Personal interviews
- Survey – face-to-face, postal method, web-based surveys
- Questionnaires
- Observation
- Demonstration (of prototypes)
- Brainstorming

# Requirement Analysis

- Common activities
  - Enumerate all requirements
  - Verify each requirement for completeness
  - Evaluate requirement
    - Technical
    - Financial
    - Timeline
  - Group requirement – functional / non-functional
  - Prioritize requirements

# Software Design

- To specify the design of software in both technical perspective and design perspective
- It describes structure of the software to be implemented, data models, interfaces between system components, and maybe the algorithms used



# Software Construction and Testing

- Creating a software (coding)
  - IDE, software dev frameworks
  - Source code control tools (Docker, Git, Github)
- Debugging and testing
  - Debugger software (most IDE provide debugging function)

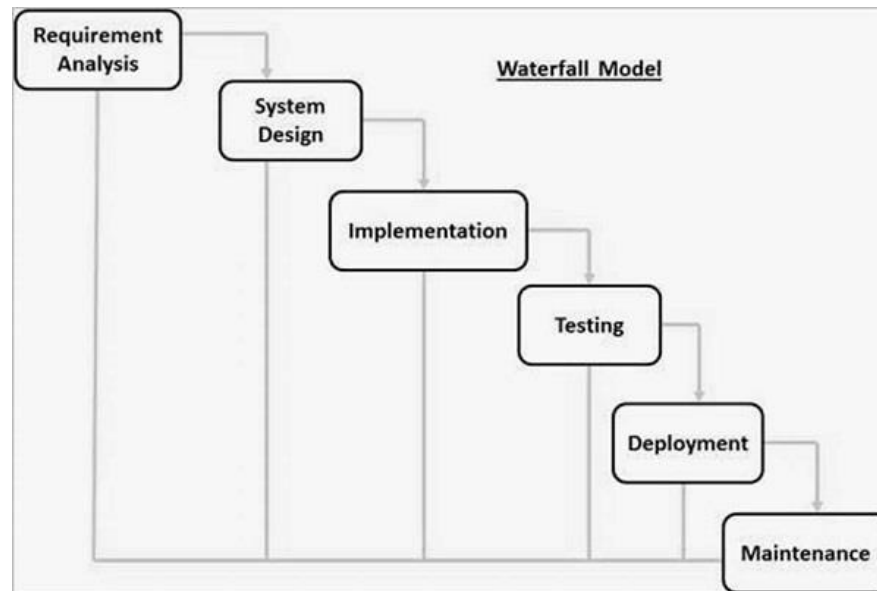
# **SOFTWARE DEVELOPMENT LIFECYCLE (SDLC)**

# Software Development Lifecycle

- Developing a software usually follows lifecycle
- There are some popular SDLCs which provide different ways of implementing software development process
  - Water fall model
  - Spiral model
  - Iterative models
  - Agile models
    - Scrum, XP, etc.

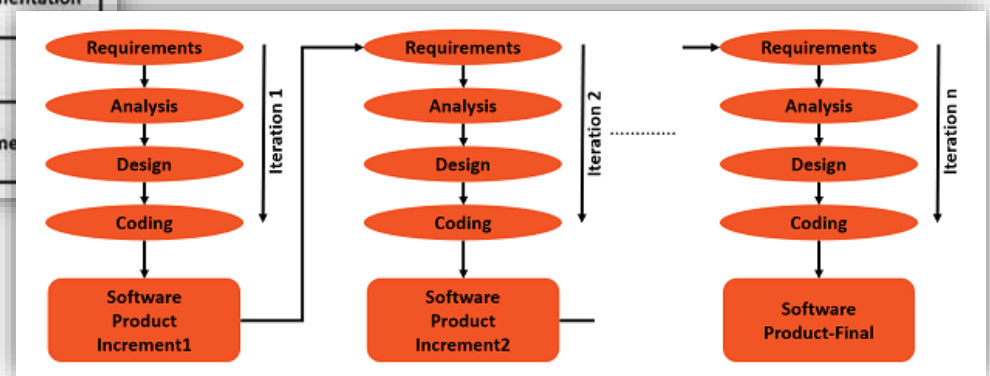
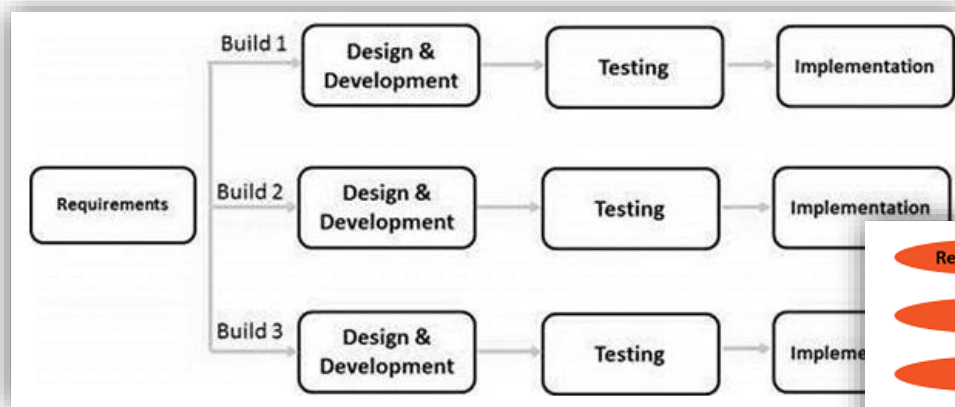
# Water Fall Model

- The whole process of software development is divided into separate phases.
- In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.



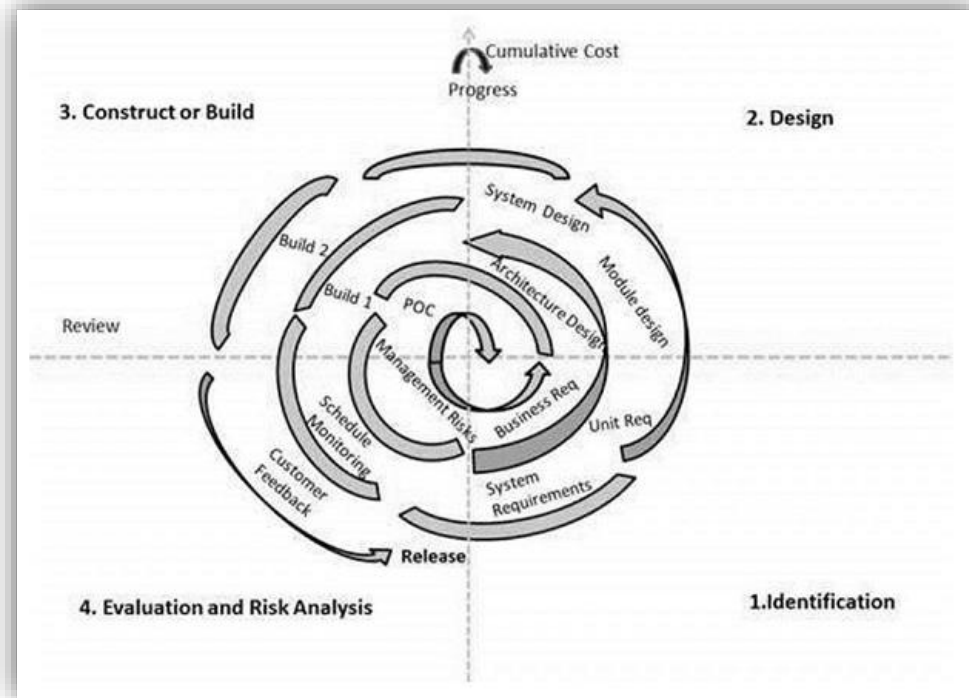
# Iterative (Incremental) Model

- A cyclic software development process (loop)
- Iterative model was extended to incremental model as well where a small set of the software requirements and iteratively enhances the evolving versions until the system is completed



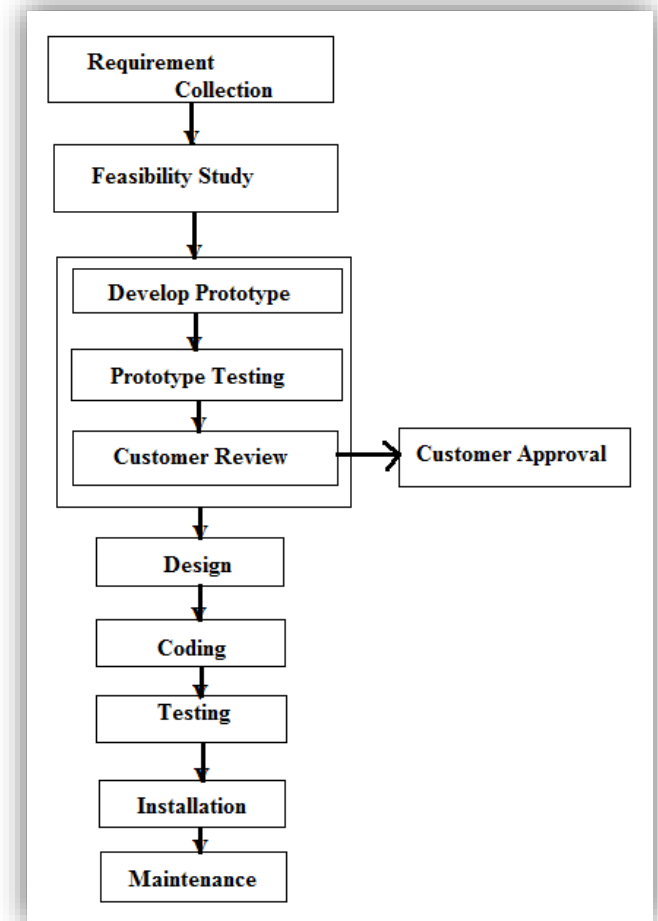
# Spiral Model

- It combines the idea of iterative development with the systematic, controlled aspects of the waterfall model
- Very high emphasis on risk analysis (risk identification is performed)



# Prototyping Model

- Before designing phase, a prototype is developed, tested, reviewed and approved by the customer
- By using this prototype, customer can understand the requirements of desired system and also the customer can get an “actual feel” of the system.

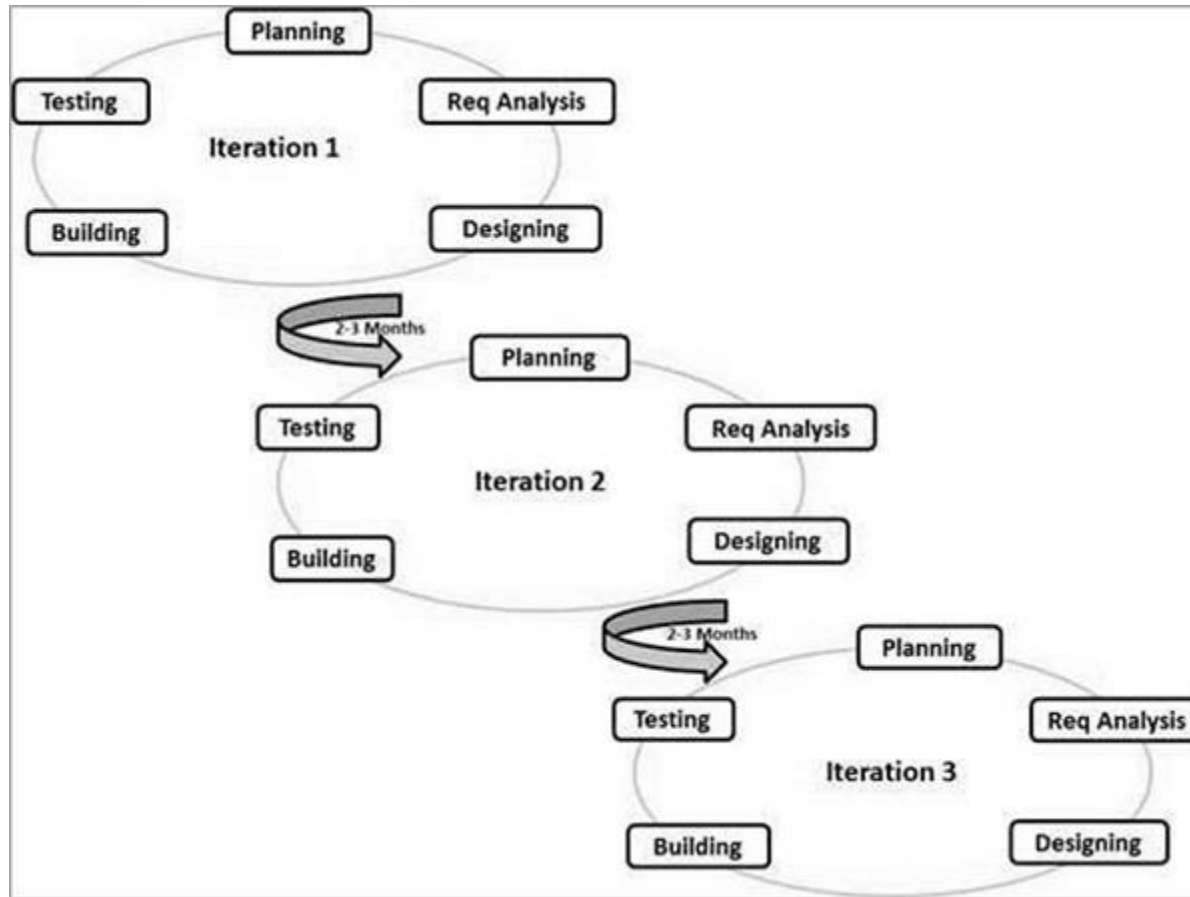


# Agile Model

- It is an adaptive software development methods
- Combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product
- Agile Methods break the product into small incremental builds
- There is feature driven development and the team adapts to the changing product requirements dynamically
- **Customer Interaction** is the backbone of this Agile methodology

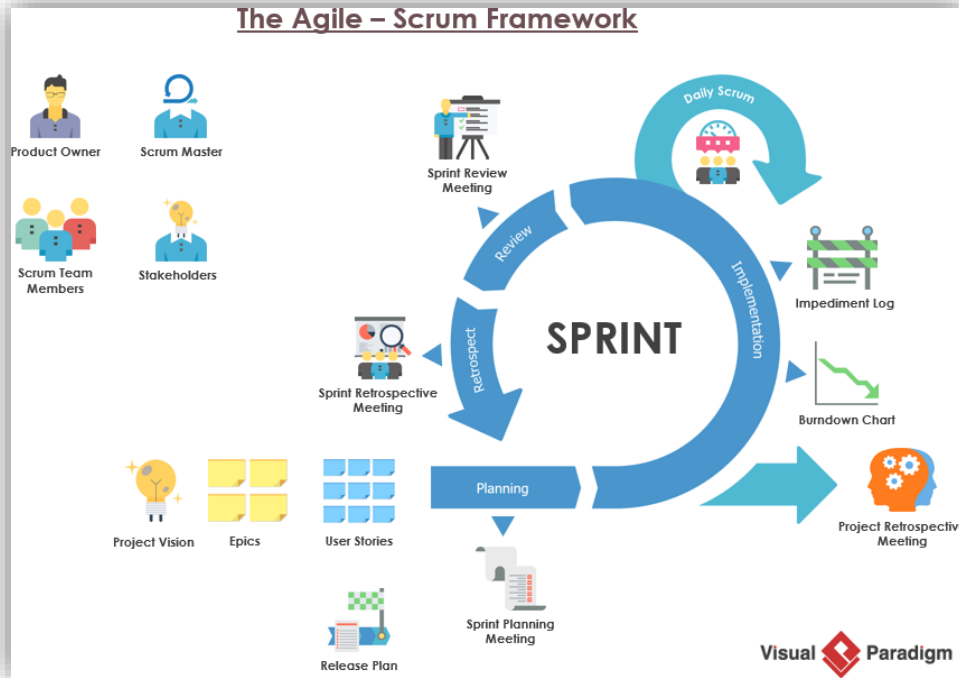


# Agile Model

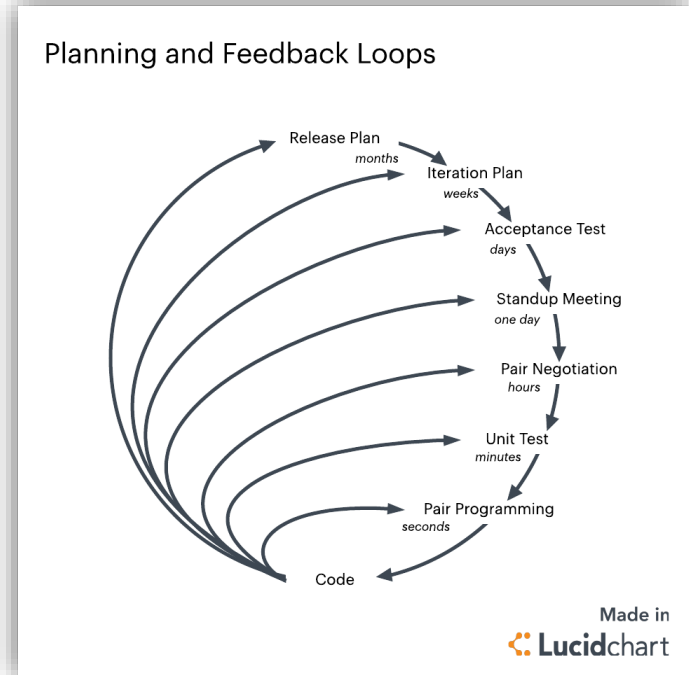


# Agile Model

- Scrum



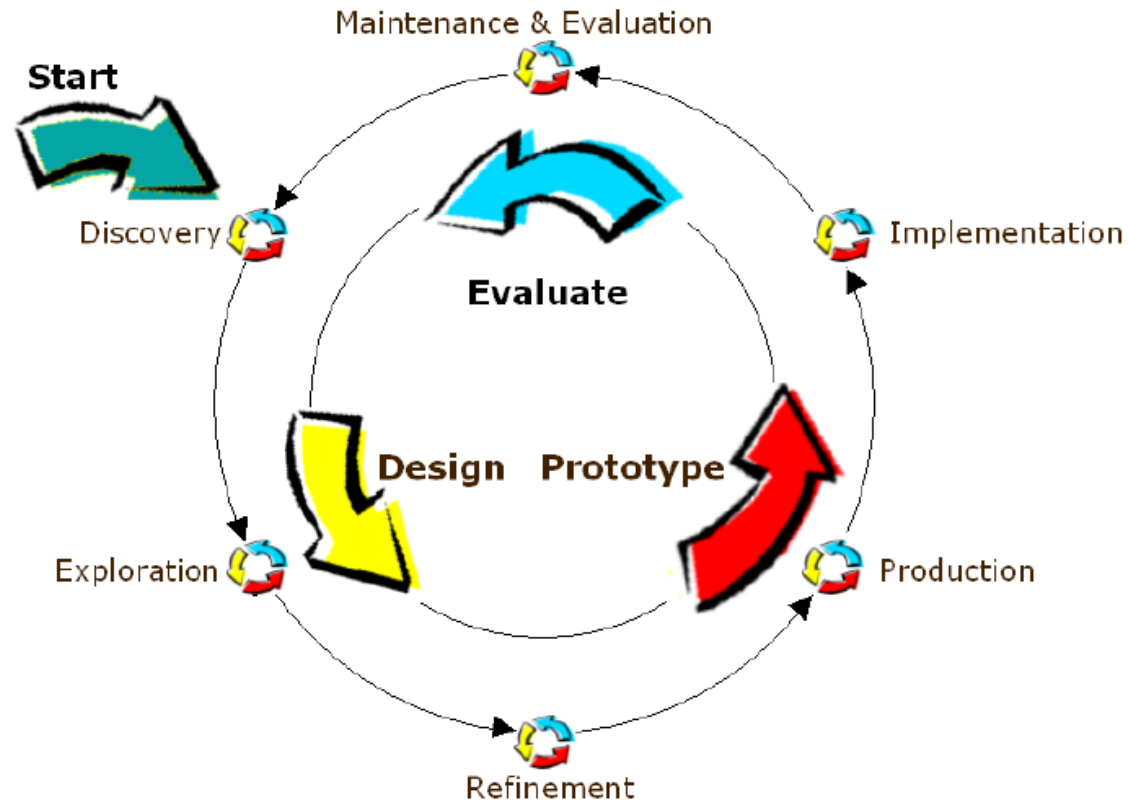
- Extreme Programming



Credit: <https://www.lucidchart.com/blog/what-is-extreme-programming>  
<https://www.visual-paradigm.com/scrum/why-scrum-difficult-to-master/>

# WEB DEVELOPMENT PROCESS

# The Website Design Process



# DISCOVERY

# Discovery

- Understand client's expectations
- Determine scope of project
- Characteristics of users
- Evaluate existing site and/or competition

# Questions to Ask

- **The first question to ask yourself is do you really need a web site?** To help you decide, ask yourself the following questions:
- **(1) Why do I want to create this web site?**
  - promote your ideas, hobbies, or beliefs
  - To advertise your company or product
  - Make loads of money really fast
  - Provide customer services and support
  - To keep your customer base informed
  - Give or sell information
  - Create an 'Extended Business Card' for your company
  - Provide internal information and services for your company

# Questions to Ask

- Before you can start deciding what content the site is going to contain you need to determine
  - Who your target audience is.
  - What age group are your users?
  - What is there skill level with the Internet?
  - Howcan I communicate effectively?
  - You also need to determine the purpose of your site. What is the site for?



# Questions to Ask

- **(2) Who is my target audience?**
  - What type of visitors do I want my site to attract? What will be their age, sex and education? Will they search for my site because we share a hobby, like the same television shows or are they looking for specific information?
- **(3) How can I communicate effectively?**
  - Now that I know who the audience will be, what is the best way to communicate with them?
- **(4) What information do I need?**
  - If you are designing a **site for a client**, you will need to know the answers to the preceding questions as well as what their vision is for this site. Do they have a logo they want you to use, do they have specific colors in mind, do they want to include phone and fax numbers on their pages? If this is your own business site, these are questions you should answer also.

# Questions to Ask

- **(5)What content (data, graphics, photos, etc.) will be included?**
  - This is the "big" job...gathering all the content that you want to include on your web site. Are you going to use photos? What kind of graphics do you want? And what information or data are you putting online?
  - Make a list of the items you think you will want to have on your website.
- **Audience analysis**
  - Audience analysis is the starting point for any project. You need to figure out your audience's demographics:
    - how old they are
    - where they work
    - what they earn
    - where they live, anything that's appropriate

# Design Tips

- Top 10 Website design tips
  - Know your audience
  - Keep webpages short
  - Limit the amount of text
  - Avoid large images
  - Use web safe colours
  - Clearly identify all links
  - Check spelling
  - Use a site map or directory page
  - Update and check all links
  - Include contact information

# EXPLORATION

# Exploration

- Visualize solutions to discovered issues
- Information & navigation design
- Early graphic design
- Select one design for development

# Research Websites

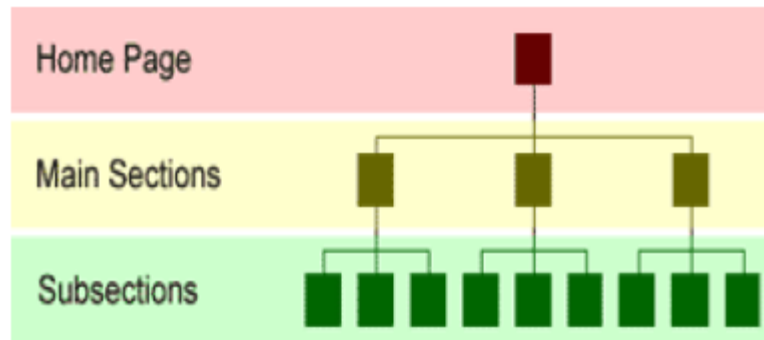
- This is a very important part in the creation of a web site, and usually very useful.
  - Spend time looking at other internet sites, particularly your competitors
  - See if you can get any ideas you can use and improve on
  - Don't be mistaken that the flashiest coolest looking web site is the best.
  - Sites with lots of animation are not always the best.
  - You must also keep in mind that not everyone has a fast Internet connection.

# Website Organisation

- There are three main elements in the organization of a web site. They are:
  - **Structure:** The form of your web site and its navigation
  - **Content:** The substance of your web site
  - **Layout:** The theme or method of presenting your web site

# Website Organisation Structure

- Site design includes the structure of your site, the different sections and navigation within those sections. It also encompasses the theme you choose.
- As the diagram shows a web site is composed of three main areas
  - The Home Page
  - The Main Sections
  - The Subsections





# Website Organisation Structure

## HomePage Do's

- Visitors first impressions
- Should tell the visitors what your site is about
- Answer questions on the 5 Ws
- Should provide index or table of contents
- Keep the home page short and to the point

## Homepage Don'ts

- Should not contain a lot of text
- Don't present your users with a huge list of links to every single page

# Website Organisation Structure

## Main Sections

- Determining and naming the main sections of your site is very important. Sections should contain material grouped according to visitor needs -in other words, these are sizeable chunks of related information.
- For example, if you are creating a site for a tourist visiting Dublin, you would want to make it easy for the visitor to find the right information. The main sections might include:

Travel

Restaurants

Pubs

Accommodation

- Once you have divided your content into main sections, decide their order of importance

# Website Organisation Structure

## Subsections

- Not all main sections necessarily need to have subsections, but most will require a further breakdown of information. It really depends on the amount of content on your site.
- When designing a new web site, keep in mind that the content will increase as you update and add information to the site. Build in room to expand as you determine your main and sub sections.
- Once you have established the home page, main sections and subsections of the web site, organise them into order of importance and note it. This is the basic layout of your web site.

# Website Organisation Navigation

There are three different navigation methods.

- **Linear navigation**-Moving in a straight line.
- **Database navigation**-Many branches from your main page.
- **Hierarchical navigation**-A completely connected website.

# Linear Navigation

- Linear navigation is used for a web site where you want the visitor to go from one step to another in a particular order.



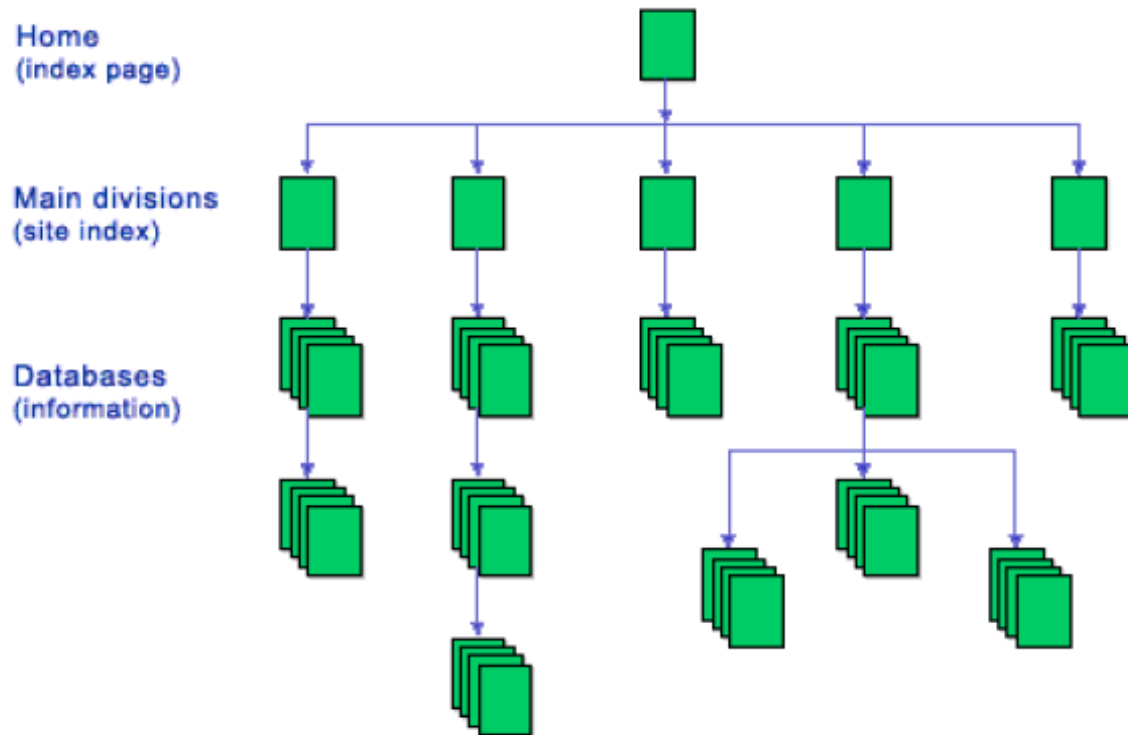
**Linear "straight line" Navigation Diagram**



**Linear "reciprocal" Navigation Diagram**

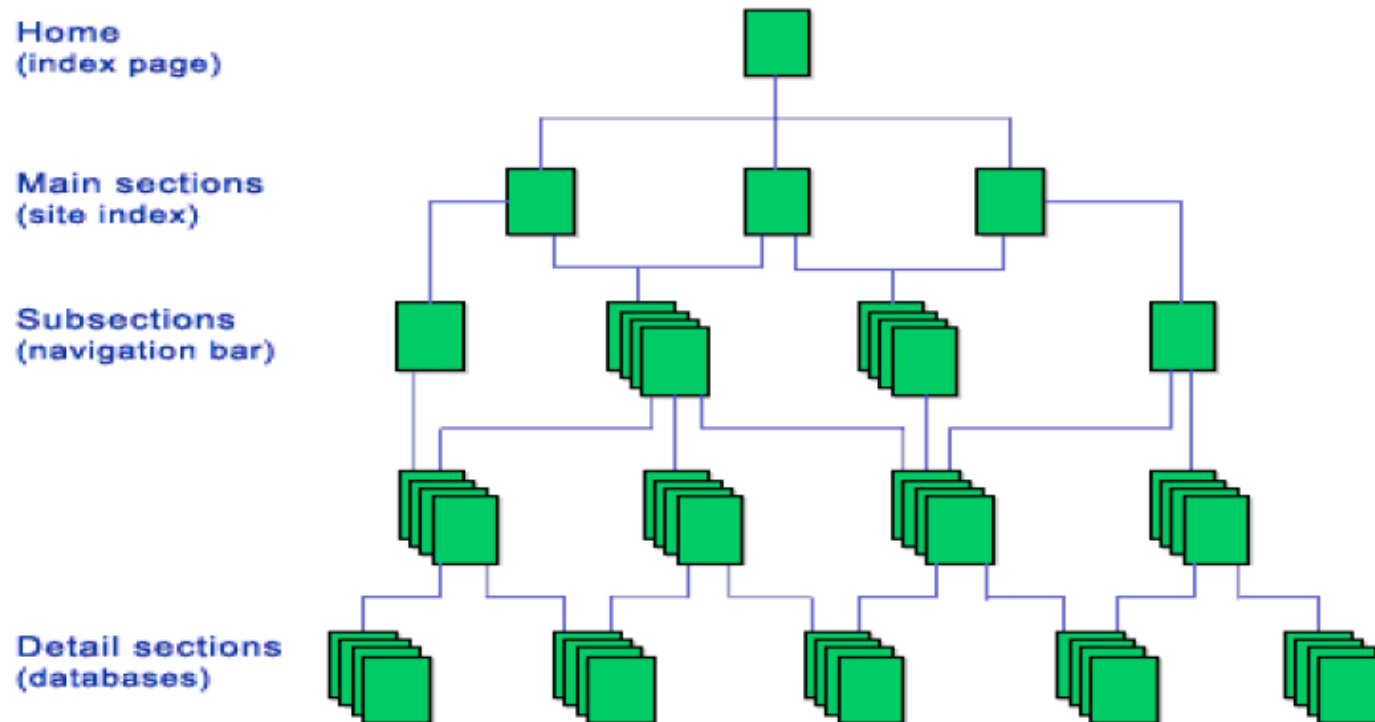
# Database Navigation

- The database or grid design is made up of multiple divisions and each division has its own structure.



# Hierarchical Navigation

- The hierarchical design goes from the general to the specific; from a home page to divisions to subdivisions.



# Website Organisation Content

- **Web Page Design** - to create the individual pages of a web site you are going to need: to be familiar with a web authoring program, or be knowledgeable in HTML coding.
- **Page Theme** – a theme can be as simple as color choice or a logo design.
- **Web Graphics** - you will need a source for graphics or be able to create your own. Again, the flexibility of knowing how to design your own graphics is preferable. PaintShopPro or PhotoShop are great programs for creating graphics, retouching photographs, designing text and building animated gifs.
- **Titles and Meta Tags** - in your HTML coding, you should always provide a descriptive title for your web pages and they should reflect what's on that particular page.

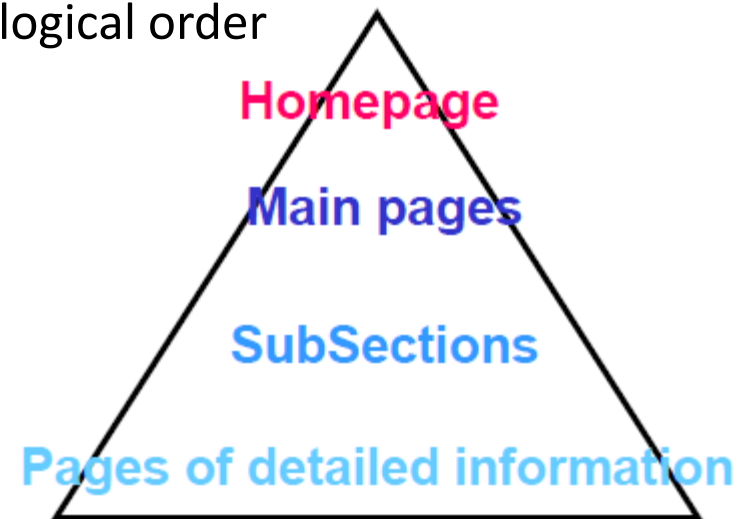


# Website Organisation Content

- **META tags** - meta tags are very important. The keywords and description provide another way for search engines to find your web site.
- **Links within your website** - the links you provide on your pages should be more than just a list of names.
  - tell visitors something about the linked site -why you like it and what they will find there.
  - You should let the site owner/web master know you are providing a link to their site. This is considered good netiquette and often results in them providing a link back to your web pages.
  - The more people you have linking to your site, the better -it helps increase your traffic.

# Website Organisation Content

- A web page should not overwhelm the visitor with unnecessary animations, flash novelties or music
- In order for visitors to understand your site you need to follow a few more rules of thumb:
  - Explain everything clearly and briefly
  - Make sure you present information in a logical order
  - Use plain English
- A web site is like a pyramid as you move down the documents increase with content.



# Website Organisation Layout

When creating a web page

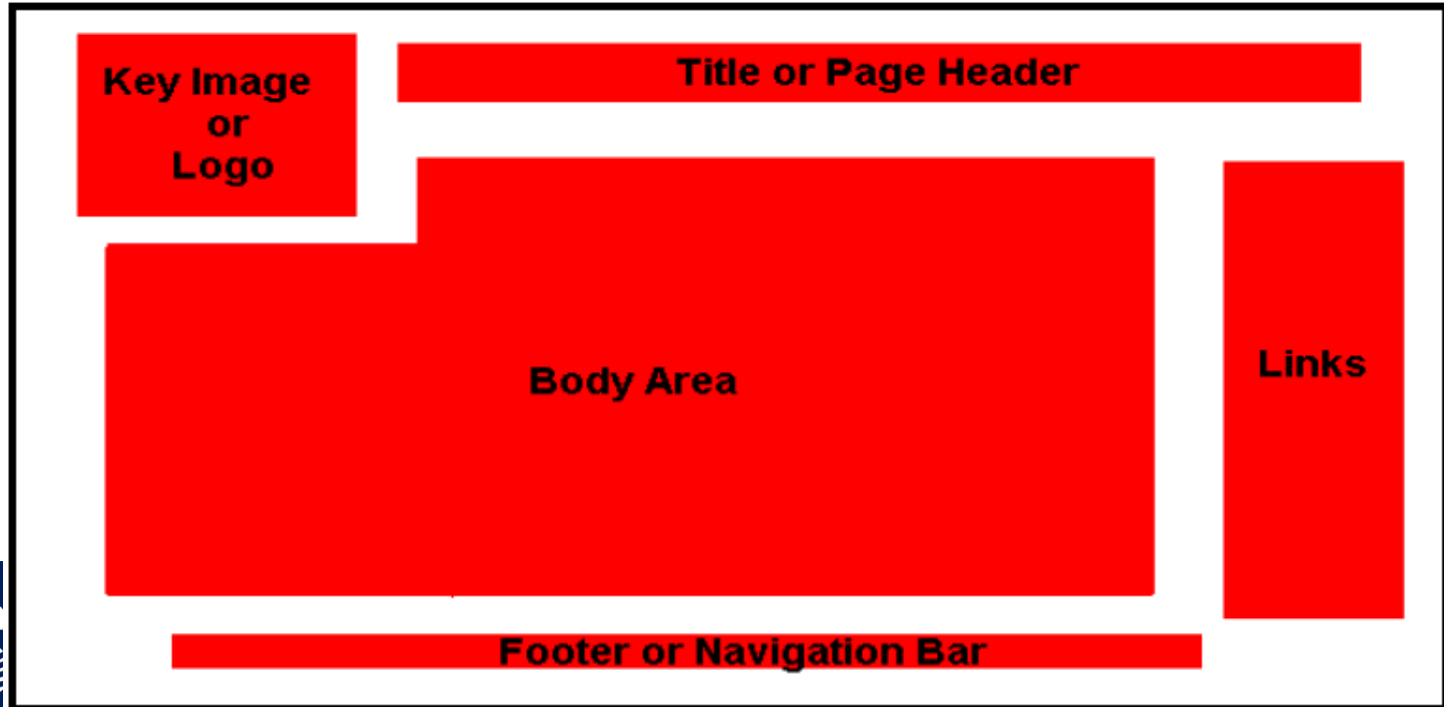
- Keep in mind that web users rarely scroll past the first screen
- Web visitors also "skim" web pages looking at the headings and focal points rather than reading paragraphs

Page layout can suffer from the following problems.

- Unbalance page layout
- No Focal Point in Page Layout
- What you should be striving for is a **Balanced and Focused Page Layout**

# Website Organisation Layout

- To produce continuity on your web site as well as enhance the ease of navigation, break what you see on the screen into regions such as header, links, footer, body, and images. Put the same information in the same regions on each page, as the following diagram shows.



# REFINEMENT

# Design Refinement

- Increasing level of detail
- Heavy emphasis on graphic design
- Iterate on design

# PRODUCTION

# Production

- Create final deliverable of design and content
- Specifications, guidelines and prototypes or mock-up
- As much detail as possible



# Site Maps

- High-level, coarse-grained view of entire site

มหาวิทยาลัยเทคโนโลยีพระจอมเกล้าธนบุรี



สำหรับ  
ผู้สนใจศึกษาต่อ
 สำหรับ  
นักศึกษาปัจจุบัน
 สำหรับ  
นักวิจัยและนักธุรกิจ
 สำหรับ  
คณาจารย์และบุคลากร

โครงสร้างเว็บไซต์

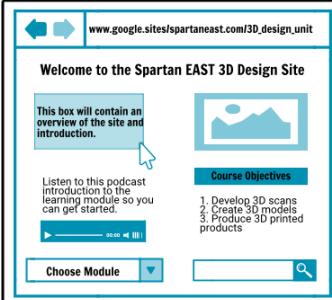
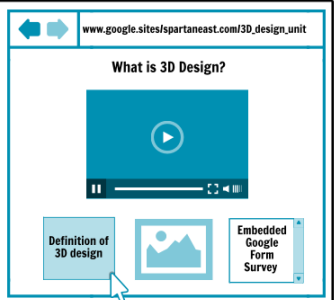
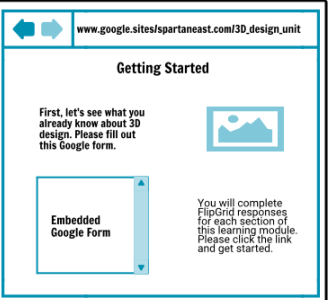
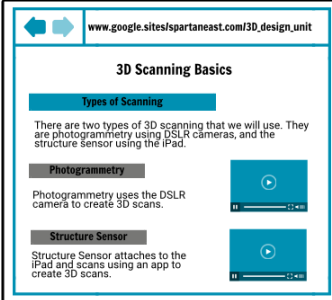
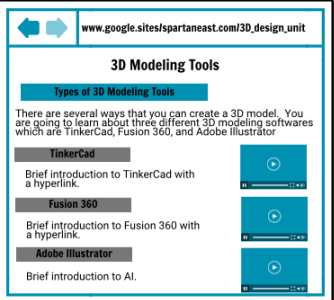
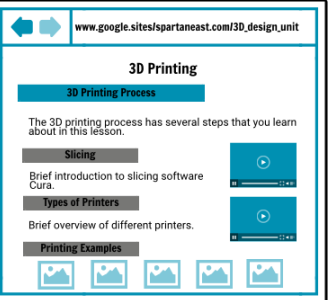
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:: [หน้าแรก](#)

| ผู้สนใจศึกษาต่อ                                                                                                                                                                                                                                                                                                                        | นักศึกษาปัจจุบัน                                                                                                                                                                                                                                                     | นักวิจัยและนักธุรกิจ                                                                                                                                                                                                    | คณาจารย์และบุคลากร                                                                                                                                                                                                                                                                                                                |
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| <b>เกี่ยวกับมหาวิทยาลัย</b> <ul style="list-style-type: none"> <li><a href="#">ข้อมูลมหาวิทยาลัย</a></li> <li><a href="#">วิสัยทัศน์</a></li> <li><a href="#">สารจากอธิการบดี</a></li> <li><a href="#">โครงสร้างการบริหาร</a></li> <li><a href="#">ผู้บริหารและคณะกรรมการ</a></li> <li><a href="#">ศักยภาพและความสำเร็จ</a></li> </ul> | <b>การสมัครเข้าศึกษา</b> <ul style="list-style-type: none"> <li><a href="#">ปริญญาตรี</a></li> <li><a href="#">บัณฑิตศึกษา</a></li> <li><a href="#">การรับบุคคลภายนอก</a></li> <li><a href="#">ทุนการศึกษา</a></li> <li><a href="#">โครงการแลกเปลี่ยน</a></li> </ul> | <b>หลักสูตร</b> <ul style="list-style-type: none"> <li><a href="#">ปริญญาตรี</a></li> <li><a href="#">บัณฑิตศึกษา</a></li> <li><a href="#">แนะนำหลักสูตรการศึกษาต่อ</a></li> <li><a href="#">ทุนการศึกษา</a></li> </ul> | <b>งานวิจัย</b> <ul style="list-style-type: none"> <li><a href="#">เกี่ยวกับงานวิจัย</a></li> <li><a href="#">ผลงานวิจัย</a></li> <li><a href="#">วารสารวิจัยและพัฒนา</a></li> <li><a href="#">สืบค้นงานวิจัย</a></li> <li><a href="#">สืบค้นเครื่องมือวิเคราะห์</a></li> <li><a href="#">มหาวิทยาลัยวิจัยแห่งชาติ</a></li> </ul> |

Credit: [https://www2.kmutt.ac.th/thai/public/public\\_map.html](https://www2.kmutt.ac.th/thai/public/public_map.html)

# Storyboards

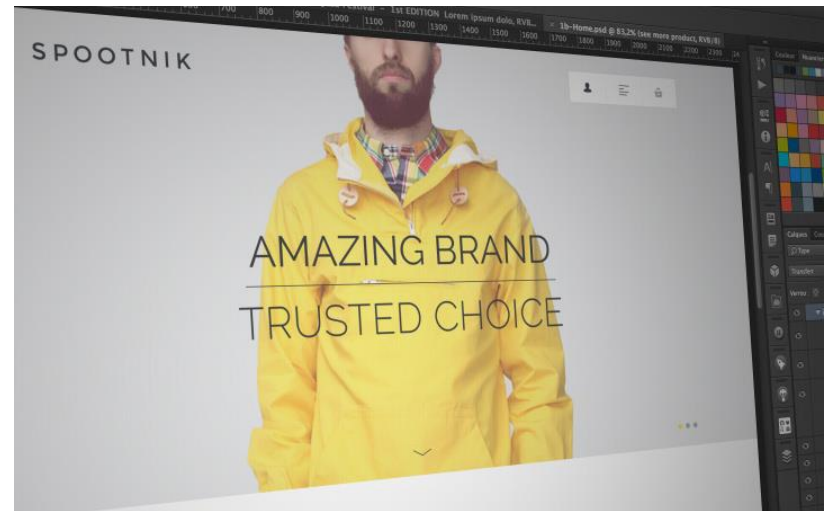
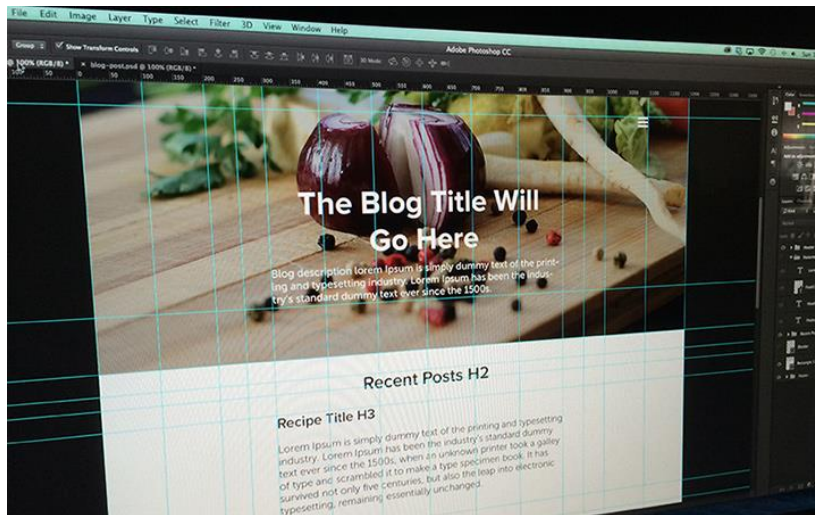
| Landing Page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Tab 1 - What is 3D Design?                                                                                                                                                                                                                                                                                                                                                                     | Tab 2 - Getting Started                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Storyboard for Landing Page. It includes a URL bar with <a href="http://www.google.sites/spartaneast.com/3D_design_unit">www.google.sites/spartaneast.com/3D_design_unit</a>, a welcome message, a placeholder for an overview image, a 'Course Objectives' section with a list: 1. Develop 3D scans, 2. Create 3D models, 3. Produce 3D printed products, a 'Listen to this podcast' section with a play button, and a 'Choose Module' dropdown menu.</p> |  <p>Storyboard for Tab 1 - What is 3D Design?. It includes the same URL bar, a 'What is 3D Design?' title, a video player placeholder, a 'Definition of 3D design' section with a placeholder image, and an 'Embedded Google Form Survey' section.</p>                                                       |  <p>Storyboard for Tab 2 - Getting Started. It includes the same URL bar, a 'Getting Started' title, a 'First, let's see what you already know about 3D design. Please fill out this Google form.' message, a placeholder for a survey, and a note about completing FlipGrid responses.</p>               |
| <p>This is the landing page for the site. There will be a brief text introduction and an audio introduction. The picture will be an example of 3D design. The course objectives will be listed on this page. There will be a dropdown menu with the link to each of the five topics. There will also be a search bar.</p>                                                                                                                                                                                                                       | <p>There will be a video about the 3D design process. Then below that there will be definition of 3D design. An image of a 3D design will be below the video, and there will be an embedded Google form survey to take information about how users feel about the site and content.</p>                                                                                                        | <p>This page will have the getting started information. There will be a survey to assess the learners' knowledge of 3D printing with an embedded Google form. There will also be an image about the assessment which have an explanation about the FlipGrid evaluation and link.</p>                                                                                                         |
| Tab 3 - 3D Scanning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Tab 4 - 3D Modeling                                                                                                                                                                                                                                                                                                                                                                            | Tab 5 - 3D Printing                                                                                                                                                                                                                                                                                                                                                                          |
|  <p>Storyboard for Tab 3 - 3D Scanning. It includes the same URL bar, a '3D Scanning Basics' title, a 'Types of Scanning' section, and two sub-sections: 'Photogrammetry' (with a video placeholder) and 'Structure Sensor' (with a video placeholder).</p>                                                                                                                                                                                                   |  <p>Storyboard for Tab 4 - 3D Modeling. It includes the same URL bar, a '3D Modeling Tools' title, a 'Types of 3D Modeling Tools' section, and three sub-sections: 'TinkerCad' (with a video placeholder), 'Fusion 360' (with a video placeholder), and 'Adobe Illustrator' (with a video placeholder).</p> |  <p>Storyboard for Tab 5 - 3D Printing. It includes the same URL bar, a '3D Printing' title, a '3D Printing Process' section, and three sub-sections: 'Slicing' (with a video placeholder), 'Types of Printers' (with a video placeholder), and 'Printing Examples' (with a placeholder for images).</p> |
| <p>This page will focus on the first objective, 3D scanning. There will be a brief explanation of 3D scanning followed by two sections one for each type. An instructional video will accompany each section.</p>                                                                                                                                                                                                                                                                                                                               | <p>This page will focus on objective two which involves 3D modeling. The design for each of the three pages for the objectives will be similar. There will be a brief introduction followed by content for each of the three modeling softwares with instructional videos for each.</p>                                                                                                        | <p>The focus of this page is objective three. The third objective focuses on the actual 3D printing. There will be a section on slicing with Cura and a section on types of printers both with instructional videos. The last section will be printing examples with photos.</p>                                                                                                             |

Create your own at Storyboard That

Credit: <https://www.storyboardthat.com/sv/storyboards/derek6949/3d-design-website>

# Mock-ups

- High-fidelity, precise representation of page



Credit: <https://www.designer-daily.com/10-examples-of-website-mockups-with-fullscreen-backgrounds-53799>

# IMPLEMENTATION

# Implementation

- This is dealing with the final checks and uploading the web site.
- Creating a index / cover page that loads quickly should be your main goal. Following are a few tips to help achieve this goal
  - Keep the page short
  - Don't make images larger than they need to be
  - Don't include background music
  - Don't include video's
  - Define width and height of all image tags
  - If image is large use thumbnail image
  - Use the same image for several of your pages, as once downloaded it will remain in the browsers cache and load instantly.

# Implementation

- To publish your website:
  - Get a domain name
  - Find a web hosting service (Free vs Private Web Hosting)
  - Upload your website

# MAINTENANCE & EVALUATION

# What Makes a Website Good?

- Webby Awards ([www.webbyawards.com](http://www.webbyawards.com))
  - “international award honoring excellence in Web design, creativity, usability and functionality”
- 500 judges, 6 criteria, 65 categories
- 6 criteria
  - Content
  - Structure & navigation
  - Visual design
  - Functionality
  - Interactivity
  - Overall experience



# Webby Awards Criteria

- Content = information provided
  - engaging, relevant
  - clear, concise
- Structure and Navigation = framework and organization of site
  - consistent, intuitive, transparent
  - facilitates forming mental model of information
- Visual Design
  - high quality, relevant, communicates visual experience, doesn't have to be trendy

# Webby Awards Criteria

- Functionality = how site works
  - loads quickly, live links, browser independent
  - experience center stage, technology invisible
  - accessibility
- Interactivity = way the site allows you to do something
  - users participate, not spectate
- Overall Experience
  - intangibles that make one stay or leave, place bookmark, sign up for newsletter, participate

# Webby Awards Criteria

- Statistical analysis applied to corpus of Webby Awards data
- Attribute most highly correlated with winning:
  - Functionality
- Least highly correlated:
  - Graphic design

# Usability

- Usability is a “quality attribute” that assesses how easy user interfaces are to use. Five components:
  - Learnability -how easy it is for users to accomplish tasks the first time they encounter the design
  - Efficiency -how quickly can users perform tasks
  - Memorability -when users return, how easily do they reestablish proficiency
  - Errors -how many errors users make, how severe, how easy to recover from errors
  - Satisfaction -how pleasant design is to use

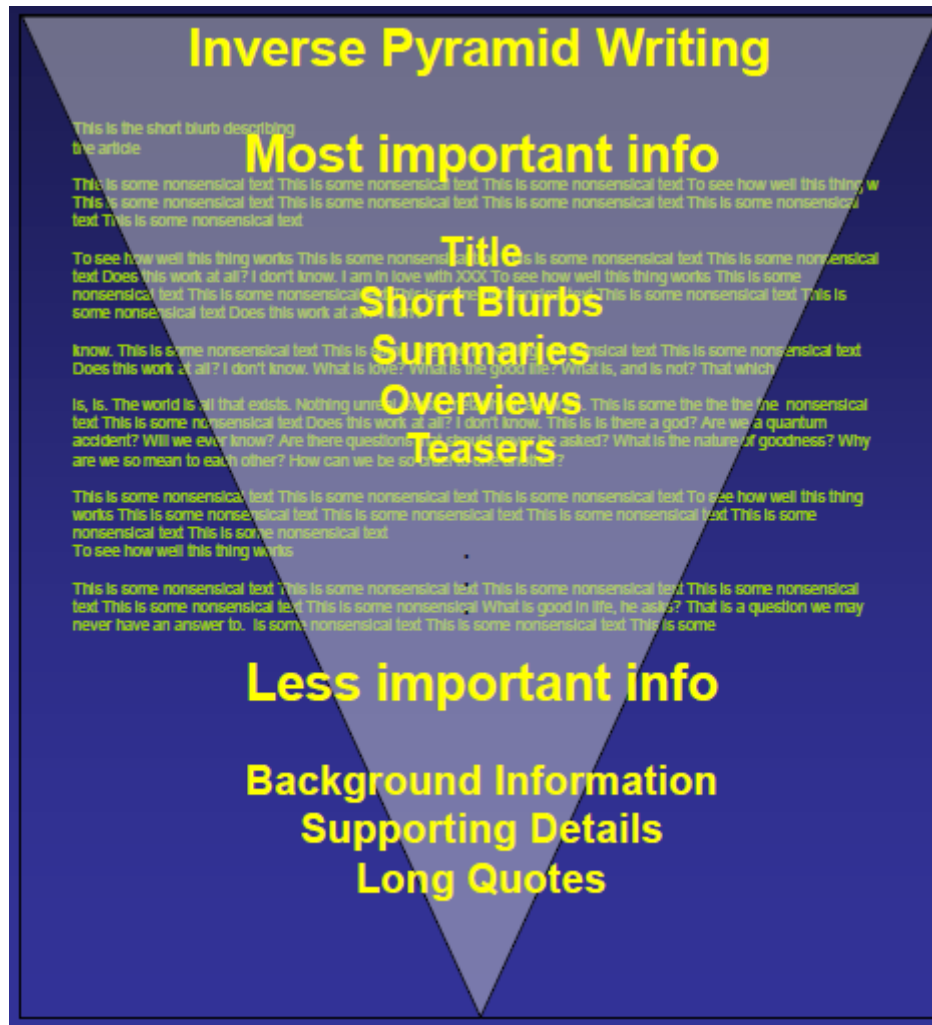
# Web Readability

- Users scan rather than read
  - Nielsen and Morkes [97] found that 79% of users always scanned any new page
- Reading from computer screens is tiring and about 25% slower than reading from paper
  - High resolution monitors, better typefaces may improve this

# Web Readability

- Use scannable text
  - Highlighted keywords
  - Meaningful sub-headings
  - Bulleted lists
- One idea per paragraph
- Inverted pyramid style of writing (conclusion first)
- Half the word count of conventional writing
- No “marketese”
- Credibility is important

# Inverse Pyramid Writing Style



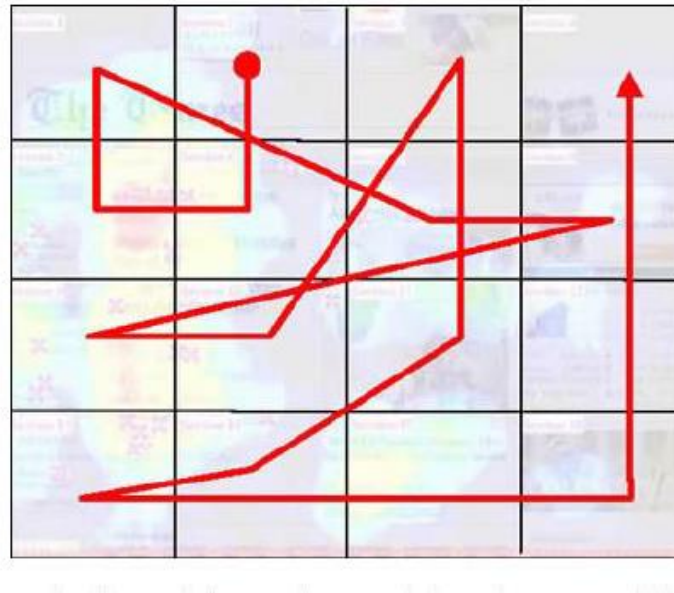
# Measuring the Usability

- Task time
  - seconds to find answers to specific content questions
- Errors
  - percentage of incorrect answers
- Memory
  - percentage of correct answers minus incorrect on post-test exam
- Time to recall site structure
  - number of seconds to draw a sitemap
- Subjective satisfaction
  - averaged 4 satisfaction-related questions



# Eye-Tracking Studies of Web Pages

- Studied where and when users look at web pages
- Eye tracking equipment (video camera looks at shape of pupil)



# How Users Look at News Web Sites (Heatmaps)



# How Users Look at News Web Sites (Heatmaps)

- Start in upper left quadrant
- Users look at text first
- Users look at the first few words of headlines
- Users read five headlines before clicking
- “Banner blindness” -users don’t look at ads or quickly look away
  - Text ads viewed more than graphic ads



# Response Times: The 3 Important Limits

- **Less than 100ms** is the limit for having user feel system is reacting instantaneously (e.g., characters typing in a text field).
- **More than 100ms but less than 1sec** is the limit for user to expect the app to process the command they have asked it to do. (e.g., process after click).
- **More than 1sec but less than 10sec** is limit for keeping user's attention focused on page (for longer delays, percent-done progress indicator should be used).
- If the action takes more than 10 sec (this is painful), you should notify users to go and have coffee...

# Other Statistics

- **Page views**-A good measure of website activity, it is the number of HTML pages served. If your visitor goes to ten pages on your site, 10 page views will be generated.
- **Visits**-A trip to your site by one person, no matter how long they stay there or how many pages they view. If they leave and come back, another visit will be generated.
- **Most popular pages**-Tells you which pages your visitors request most. Helps you learn what works!
- **Average number of pages per visit**-If this is low, you know your site sucks! You need to do some revisions. But don't give up! Just keep working at it until that number goes up!
- **Referring pages**-This is an important one because it tells you where your visitors link FROM, who has links to your site, and what search engines you're listed with properly.
- **Browsers used**-Tells you what browsers people are using and helps you know how to design your web pages so that people can use them!

# Exercise 2.1

- Discuss pro and con of each SDLC approach. Submit as a pdf file.